CLAUDE CERTIFIED ARCHITECT — PROFESSIONAL

Objective 21: Design Evaluation Datasets and Frameworks

Exam objective: Design evaluation datasets and frameworks, including building representative datasets, using real-world examples, edge cases, and adversarial test cases, creating regression datasets, using code-based and model-based graders, incorporating human evaluation, evaluating multi-turn conversations, and calibrating model graders against human judgment.

Q1 **An architect at a logistics company is building the first evaluation dataset for a Claude-based shipment-exception-handling agent. The team has only synthetic examples written by engineers. What is the most important improvement to make before launch?**

A. Add more synthetic examples generated by the same engineers

B. Incorporate real anonymized production transcripts representative of actual customer issues

C. Increase the dataset size without changing its source

D. Remove all edge cases to simplify scoring

CORRECT ANSWER **B**

WHY Real-world examples capture the actual distribution and messiness of production inputs that synthetic data written by engineers tends to miss, making the dataset genuinely representative. Simply enlarging or simplifying a synthetic-only set does not fix the representativeness gap, and removing edge cases reduces coverage.

Q2 **A healthcare provider's clinical-notes summarization evaluation dataset performs well on typical visit notes but has never been tested against malformed, truncated, or contradictory input notes. What category of test case is missing?**

A Regression test cases

B Edge cases

C Model-based grader calibration cases

D A/B testing cases

CORRECT ANSWER **B**

WHY Malformed, truncated, or contradictory inputs are edge cases — unusual but plausible inputs that stress boundary conditions the typical-case dataset doesn't cover. Regression cases track behavior over time, calibration cases validate graders, and A/B cases compare variants, none of which address input-coverage gaps.

Q3 **A bank's fraud-detection assistant must be evaluated for resistance to users attempting to manipulate it into approving suspicious transactions through crafted prompts. Which dataset component specifically addresses this risk?**

A Regression dataset

B Adversarial test cases (e.g., prompt injection and jailbreak attempts)

C Real-world representative examples only

D Latency benchmark cases

CORRECT ANSWER **B**

WHY Adversarial test cases are specifically designed to probe manipulation attempts such as prompt injection or jailbreaks, which is exactly the fraud-approval manipulation risk described. Regression and representative datasets validate normal behavior, and latency cases measure speed, not robustness to attack.

Q4 After fixing a bug where a Claude-based legal-document redaction tool failed to redact Social Security numbers in a specific format, the architect wants to ensure this exact failure never silently reappears in future releases. What should be added to the evaluation suite?

A. A new adversarial prompt category

B. A regression test case capturing the exact failure scenario

C. A higher sampling temperature for future runs

D. A new human evaluation rubric only

CORRECT ANSWER B

WHY Regression datasets exist precisely to lock in fixes for previously identified failures so that future changes are automatically checked against them. Adversarial categories address attack robustness, not confirmed bug fixes, and temperature/rubric changes don't institutionalize the fix as a repeatable test.

Q5 A retail company wants to automatically check thousands of outputs per day from its product-FAQ agent for whether responses contain a required disclaimer sentence verbatim. Which grading approach is most efficient and appropriate?

A. Human evaluation for every output

B. Code-based grader (exact string/regex match)

C. Model-based grader with open-ended judgment

D. User satisfaction survey

CORRECT ANSWER B

WHY Checking for a verbatim, well-defined string is a deterministic pattern-matching task ideally suited to a cheap, fast, and reliable code-based grader using exact match or regex. Human and model-based graders are costlier and less precise for simple deterministic checks, and satisfaction surveys don't verify specific content.

Q6 **An insurance claims assistant's responses must be judged on nuanced qualities like "empathy toward distressed claimants" and "appropriate tone," which cannot be captured by simple string matching. Which grading approach is best suited here?**

A Code-based grader using keyword matching

B Model-based grader (an LLM judge) with a clear rubric

C Latency-based grader

D Token-count-based grader

CORRECT ANSWER **B**

WHY Subjective, nuanced qualities like empathy and tone require judgment beyond pattern matching, which a model-based grader (LLM-as-judge) using a well-defined rubric can approximate at scale. Keyword matching, latency, and token-count graders cannot assess qualitative tone or empathy.

Q7 **A SaaS company has deployed an LLM-based grader to score its support-agent's helpfulness at scale. Before trusting its scores for release-gating decisions, what must the architect do?**

A Deploy the grader to production immediately since LLMs are inherently reliable judges

B Calibrate the model-based grader against a sample of human-labeled ground truth

C Replace the grader with a code-based grader permanently

D Use the grader only for adversarial test cases

CORRECT ANSWER **B**

WHY Model-based graders must be validated (calibrated) against human judgments on a representative sample to confirm their scores correlate well with actual quality before being trusted for high-stakes decisions like release gating. Skipping calibration risks systematic grader bias going undetected, and the other options don't address the trust/validation gap.

Q8 **A telecom company's virtual assistant handles conversations spanning 8–10 turns where context (e.g., a previously stated account number) must be retained and used correctly later. What evaluation framework limitation would an architect need to specifically address?**

A Single-turn accuracy metrics are sufficient since each turn is independent

B The framework must evaluate multi-turn conversations holistically, including context retention across turns

C Only the final turn needs to be evaluated

D Multi-turn evaluation is unnecessary if latency is acceptable

CORRECT ANSWER B

WHY Multi-turn agents can fail by losing or misusing context established earlier in a conversation, a failure mode invisible to single-turn or final-turn-only evaluation; the framework must score the full conversation trajectory. The other options ignore cross-turn dependencies entirely.

Q9 **A government agency building an evaluation dataset for a citizen-services chatbot wants to ensure the dataset reflects the true diversity of citizen queries, not just the cases engineers anticipated. What is the best practice to achieve this?**

A Rely solely on internally brainstormed hypothetical scenarios

B Sample and curate real (anonymized/de-identified) interaction logs alongside synthetic edge cases

C Use only adversarial prompts written by a red team

D Use a fixed dataset created once and never updated

CORRECT ANSWER B

WHY Combining real anonymized interaction data with deliberately constructed edge cases gives the broadest, most representative coverage of both typical and unusual real-world usage. Relying only on internal brainstorming, only adversarial prompts, or a static unchanging dataset each leaves significant representativeness gaps.

Q10 **A manufacturing firm's architect notices that their model-based grader consistently rates verbose answers higher than concise-but-correct ones, even when human reviewers prefer the concise answers. What does this indicate and what should be done?**

A The grader is fine; verbosity is a legitimate proxy for quality

B The grader has a bias not aligned with human judgment and needs recalibration or rubric revision

C The dataset needs more adversarial cases

D The issue is a latency problem, not a grading problem

CORRECT ANSWER B

WHY A systematic divergence between model-grader scores and human judgment reveals a miscalibrated or biased grader (e.g., favoring length over correctness), which requires rubric revision or recalibration before the grader can be trusted. It is not a latency or adversarial-coverage issue.

Q11 **An architect at a law firm is designing an evaluation framework for a contract-clause classifier and wants to combine the speed of automated grading with the nuance of human judgment, given limited reviewer budget. What is the most practical approach?**

A Use human evaluation for 100% of all outputs indefinitely

B Use code/model-based graders for bulk scoring, with periodic human evaluation on a sampled subset for calibration and spot-checking

C Use only code-based graders and eliminate human review entirely

D Use only human evaluation for a single one-time assessment, then stop evaluating

CORRECT ANSWER B

WHY A hybrid approach — automated graders for scale plus periodic human sampling for calibration and quality assurance — balances cost and rigor, which is standard practice for sustainable evaluation at scale. Pure human review is unsustainable at volume, pure automation risks undetected drift, and a one-time assessment doesn't support ongoing monitoring.

Q12 **A logistics company wants its evaluation dataset for a route-optimization assistant to include test cases where the input deliberately contains conflicting constraints (e.g., impossible delivery windows). What is the primary purpose of including such cases?**

A To test the model's latency under heavy load

B To test the model's graceful handling of edge cases and ambiguous/conflicting input

C To test token pricing accuracy

D To replace the need for regression testing

CORRECT ANSWER **B**

WHY Deliberately conflicting or impossible constraints are a form of edge case designed to test whether the system detects the conflict and responds gracefully (e.g., flags the issue) rather than failing silently or producing a nonsensical plan. Latency, pricing, and regression testing are unrelated concerns.

Q13 A financial-services firm's evaluation framework for an investment-research assistant includes a "golden dataset" of 200 curated examples with expert-verified correct answers. Six months later, the product has evolved to support five new query types not in the golden set. What should the architect do?

A Continue using the original 200 examples indefinitely since they were expert-verified

B Expand and evolve the evaluation dataset to include representative examples of the new query types

C Discard the golden dataset entirely and rely only on user satisfaction scores

D Freeze the dataset to maintain consistent historical comparisons forever

CORRECT ANSWER **B**

WHY Evaluation datasets must evolve alongside the product to remain representative; failing to add coverage for new capabilities leaves significant functionality unevaluated. Freezing the dataset forever or discarding it in favor of subjective satisfaction scores both undermine representativeness and rigor.

Q14 An architect building adversarial test cases for a customer-facing retail chatbot wants to specifically probe whether a malicious user can extract the system prompt or internal instructions. Which category of test best fits this goal?

A Edge case for malformed input

B Prompt injection / system-prompt-extraction adversarial test

C Regression test for a previously fixed bug

D Multi-turn context retention test

CORRECT ANSWER **B**

WHY Attempts to extract system prompts or internal instructions are a well-known adversarial/prompt-injection attack pattern that must be explicitly tested, distinct from malformed-input edge cases, bug regressions, or context-retention checks.

Q15 **A pharmaceutical company's evaluation team disagrees on whether to prioritize a large automatically-generated dataset (10,000 synthetic examples) or a smaller, carefully curated set (300 examples) with expert clinical review for a drug-interaction-checking assistant. What is the most defensible architectural recommendation?**

A Always prefer the larger synthetic dataset because sample size alone determines evaluation quality

B **Prioritize the smaller expert-curated set for high-stakes correctness, supplemented by the larger synthetic set for broader coverage and regression monitoring**

C Use only the synthetic dataset since human expert time is too costly to justify

D Discard both and rely solely on production user complaints

CORRECT ANSWER **B**

WHY For high-stakes domains like drug interactions, expert-verified correctness is irreplaceable for core quality gating, while a large synthetic set still adds value for breadth and regression coverage — a complementary combination outperforms either alone. Raw sample size without quality control, or waiting for user complaints, is insufficient for safety-critical evaluation.

Q16 An architect calibrating a model-based grader for a customer-service assistant writes a rubric that simply says "score 1-10 for helpfulness" with no further guidance, and finds two human reviewers scoring the same transcript a 6 and a 9. What is the most effective fix to the rubric itself?

A Remove the rubric entirely and let the grader decide freely

B Add concrete, example-anchored descriptions of what output looks like at each score point (or key points), so graders and humans share a common standard

C Switch exclusively to code-based grading since human disagreement proves rubrics don't work

D Lower the scoring scale to a binary pass/fail to avoid any further disagreement

CORRECT ANSWER **B**

WHY Score anchors — concrete descriptions or examples of what distinguishes adjacent points on the scale — give both human reviewers and model-based graders a shared, reproducible standard, directly addressing the ambiguity that caused the 6-vs-9 disagreement. Removing guidance, abandoning rubrics for pure keyword matching, or collapsing to binary scoring would lose the nuance the use case requires rather than fix the underlying ambiguity.

Q17 **A retail company builds an evaluation set for its product-support agent by deliberately rebalancing categories so that rare-but-important query types (e.g., warranty disputes) each make up an equal 10% share, even though they represent under 1% of real production traffic. The resulting aggregate score looks strong. What caution should the architect apply when reporting this score?**

A None; a rebalanced set is always preferable to one that mirrors production

B **The aggregate score should be reported alongside the dataset's category distribution, since a rebalanced set describes a different (non-production) population and the overall number can mislead stakeholders about real-world performance**

C Rebalancing eliminates the need for any further regression or adversarial testing

D The rebalanced score should replace production monitoring entirely going forward

CORRECT ANSWER **B**

WHY Deliberately rebalancing a dataset is a legitimate way to ensure rare categories get adequate coverage, but the resulting aggregate score reflects a synthetic population mix, not actual traffic — so the category distribution must be documented and considered when interpreting or communicating the overall number. Treating the rebalanced score as a direct proxy for production performance, or as a substitute for regression/adversarial testing or live monitoring, misrepresents what the number actually measures. --